Shreya Nasa

@ shrnasa011@gmail.com West Lafayette, IN LinkedIn Homepage

PhD student in theoretical computer science working on **coding theory** (interactive communication, error resilience under bit-flip and erasure noise), **sublinear algorithms and property testing**, and **extremal combinatorics** (hypergraph containers, Shannon capacity of graphs). Three years at Goldman Sachs on a large-scale **knowledge graph** and its **data pipelines**.

Education

MS + PhD in Computer Science — **Purdue University** — Aug 2022–May 2028 (exp.) — West Lafayette, IN — GPA: 4.0/4.0

B.Tech. in ECE with Honors — **IIT Roorkee** — Aug 2015–May 2019 — Roorkee, India — CGPA: 9.13/10

Research Experience

Graduate Researcher — **Purdue University** — Mar 2023–Present — West Lafayette, IN

Advisors: Prof. Elena Grigorescu and Prof. Kent Quanrud

- **Sublinear algorithms / property testing:** Used hypergraph containers to show that the independent-set property in q-uniform hypergraphs can be tested with $\widetilde{O}\big(q\rho^{2q-3}/(\varepsilon^2((q-2)!)^2)\big)$ samples, which is optimal in ε and exponentially better in q *(RANDOM 2026)*.
- **Coding theory with feedback and erasures:** Characterized the limits of error resilience and the maximal asymptotic rate of communication with perfect feedback when the channel suffers both bit flips and erasures *(ISIT 2025)*.
- **Interactive error-correcting codes:** Designed new adversarial attack strategies that give an upper bound of $7/24$ on the tolerable error fraction for interactive ECCs with separate corruption budgets *(Allerton 2025)*.
- **Extremal graph theory:** Ongoing project on the Shannon (zero-error) capacity $\Theta(G)$ of graphs, mapping known bounds, techniques, and open problems.

Publications

Testing the Independent Set Property in Hypergraphs — *RANDOM 2026*
*Elena Grigorescu, **Shreya Nasa**, Cameron Seth**

Communication with Perfect Feedback for Bit Flips and Erasures — *ISIT 2025*
*Elena Grigorescu, **Shreya Nasa**, Maoyuan Song**

An Upper Bound for Interactive ECCs with Separate Budgets — *Allerton 2025*
***Shreya Nasa**, Maoyuan Song**

* Authors listed alphabetically; all contributed equally. **I presented all three papers at the respective conferences.**

Teaching Experience

Graduate Teaching Assistant — **Purdue University** — 2024–Present — West Lafayette, IN

- **Head TA**, graduate **CS 58000: Algorithm Design** (Fall 2024): led the design and grading of all assignments and exams.
- TA, graduate **CS 57800: Statistical Machine Learning** (Spring 2024, Fall 2026) and **CS 18000: Problem Solving and Object-Oriented Programming** (Spring 2026).

Industry Experience

Associate / Analyst — **Goldman Sachs** — Jun 2019–Jun 2022 — Bengaluru, India

- Managed and extended the firm's **knowledge graph** (HBase) for the Compliance division's Relationship Analytics and Engineering team; integrated firm-wide e-communication data and graph-derived attributes with the infrastructure and inference teams.
- Optimized **MapReduce** jobs in the communication-data pipeline, cutting end-to-end runtime **3**$\times$ (24 hours to 8 hours).
- Built a neural model of relationship strength between graph vertices, raising F1 from 0.40 (logistic regression) to 0.89.
- Onboarded and mentored junior team members; panelist at mentoring and outreach events for prospective hires.

Skills & Interests

Theory: coding theory, property testing and sublinear algorithms, extremal combinatorics, graph theory, machine learning
Programming: Python, C/C++, Java, SQL, MATLAB; PyTorch, TensorFlow, scikit-learn; MapReduce, HBase, Git

Selected Achievements

Google Code Jam to I/O for Women: ranked 9th (2019) and 69th (2018).
Academic awards, IIT Roorkee: Prof. Ranvir Singh Shastry Cash Prize (second-highest CGPA among B.Tech. women); Rai Singh Jain Cash Prize (highest CGPA among B.Tech. women in CSE/ECE/Electrical, 2016 and 2017–18).